

ECON 300 – Labour Economics

Solutions: Final Exam Numerical Problems

Question 1: Labour Supply, Policy, and Unemployment

Johnny has:

$$T = 100, \quad h = T - L, \quad w = 20, \quad V = 200$$

where L is leisure, h is hours worked, and V is non-labour income.

(a) **Budget constraint**

Income is labour earnings plus non-labour income:

$$C = wh + V$$

Since:

$$h = T - L = 100 - L$$

Substitute into the income equation:

$$C = 20(100 - L) + 200$$

$$C = 2000 - 20L + 200$$

$$\boxed{C = 2200 - 20L}$$

Income intercept:

Set $L = 0$:

$$C = 2200 - 20(0) = 2200$$

$$\boxed{2200}$$

Leisure intercept:

Set hours worked equal to zero, so $L = T = 100$.

$$\boxed{100}$$

Slope:

From

$$C = 2200 - 20L$$

the slope is:

$$\boxed{-20}$$

Interpretation: one more hour of leisure reduces income by \$20.

(b) **Guaranteed income program**

Given:

$$G = 600, \quad t = 0.5$$

Disposable income while benefits are positive is:

$$C = G + (1 - t)wh$$

Substitute:

$$C = 600 + (1 - 0.5)(20)h$$

$$C = 600 + 0.5(20)h$$

$$\boxed{C = 600 + 10h}$$

Effective net wage:

$$(1 - t)w = 0.5(20) = 10$$

$$\boxed{\$10}$$

Break-even level of weekly earnings:

Benefits go to zero when:

$$G = twh$$

or:

$$wh = \frac{G}{t}$$

$$wh = \frac{600}{0.5} = 1200$$

$$\boxed{\$1200}$$

Break-even hours worked:

Since $w = 20$:

$$20h = 1200$$

$$h = 60$$

60 hours

(c) **Change in labour supply**

Initially Johny works:

40 hours

After the program:

30 hours

Change in labour supply:

$$30 - 40 = -10$$

-10 hours

So labour supply falls by 10 hours.

Interpretation:

This is consistent with:

- an **income effect**: the guaranteed income makes Johny richer even without working, so he may choose more leisure;
- a **substitution effect**: the effective wage falls from \$20 to \$10, so leisure becomes cheaper relative to work.

Both effects reduce labour supply.

(d) **Labour force, unemployed, employed**

Working-age population:

5,000,000

LFPR:

70%

Unemployment rate:

6%

Labour force:

$$0.70(5,000,000) = 3,500,000$$

$$\boxed{3,500,000}$$

Unemployed workers:

$$0.06(3,500,000) = 210,000$$

$$\boxed{210,000}$$

Employed workers:

$$3,500,000 - 210,000 = 3,290,000$$

$$\boxed{3,290,000}$$

(e) **New unemployment and labour force**

Initial unemployed:

$$210,000$$

Initial labour force:

$$3,500,000$$

Changes:

- 60,000 unemployed find jobs $\Rightarrow$ unemployment falls
- 40,000 employed lose jobs and search $\Rightarrow$ unemployment rises
- 30,000 workers leave the labour force

New unemployed:

Start with initial unemployed:

$$210,000$$

Subtract those who find jobs:

$$210,000 - 60,000 = 150,000$$

Add those who lose jobs:

$$150,000 + 40,000 = 190,000$$

$$\boxed{190,000}$$

New labour force:

The 30,000 workers leaving the labour force reduce the labour force:

$$3,500,000 - 30,000 = 3,470,000$$

$$\boxed{3,470,000}$$

New unemployment rate:

$$\frac{190,000}{3,470,000} \times 100 \approx 5.48\%$$

$$\boxed{5.48\%}$$

Interpretation:

Even though some workers lost jobs, the unemployment rate falls because many unemployed found jobs and some workers left the labour force.

Question 2: Labour Demand, Monopsony, Immigration, and Human Capital

Demand:

$$W = 50 - 0.5N$$

Supply:

$$W = 10 + 0.5N$$

(a) **Competitive equilibrium**

Set demand equal to supply:

$$50 - 0.5N = 10 + 0.5N$$

$$40 = N$$

$$\boxed{N = 40}$$

Substitute into either curve:

$$W = 10 + 0.5(40) = 10 + 20 = 30$$

$$\boxed{W = 30}$$

So the competitive equilibrium is:

$$\boxed{(N, W) = (40, 30)}$$

(b) **Monopsony hiring 20 workers**

If the monopsonist hires:

$$N = 20$$

Then the wage paid comes from the supply curve:

$$W = 10 + 0.5(20) = 10 + 10 = 20$$

$$\boxed{W = 20}$$

Compare this to competitive wage:

$$20 < 30$$

Monopsony wage is \$10 lower than the competitive wage

Explanation:

A monopsonist has labour market power and faces an upward-sloping supply curve. It can lower wages by restricting employment below the competitive level.

(c) **Minimum wage of \$30**

Workers willing to work at \$30:

Use the supply curve:

$$30 = 10 + 0.5N$$

$$20 = 0.5N$$

$$N = 40$$

$$\boxed{40}$$

Workers demanded at \$30:

Use the demand curve:

$$30 = 50 - 0.5N$$

$$0.5N = 20$$

$$N = 40$$

$$\boxed{40}$$

Actual employment:

Employment is the smaller of labour supply and labour demand:

$$\boxed{40}$$

So:

$$\boxed{\text{Actual employment} = 40}$$

(d) **Immigration shares**

Total immigrants:

$$400,000$$

Economic share:

$$\frac{200,000}{400,000} \times 100 = 50\%$$

$$\boxed{50\%}$$

Family share:

$$\frac{120,000}{400,000} \times 100 = 30\%$$

$$\boxed{30\%}$$

Refugee share:

$$\frac{80,000}{400,000} \times 100 = 20\%$$

$$\boxed{20\%}$$

If economic immigrants rise by 10%:

$$200,000(1.10) = 220,000$$

New total:

$$220,000 + 120,000 + 80,000 = 420,000$$

$$\boxed{420,000}$$

(e) **Human capital investment**

Given:

$$A = 3000, \quad r = 0.05, \quad T = 8$$

Present value of benefits:

$$PV = 3000 \left(\frac{1 - (1.05)^{-8}}{0.05} \right)$$

First compute:

$$(1.05)^{-8} \approx 0.67684$$

So:

$$PV = 3000 \left(\frac{1 - 0.67684}{0.05} \right)$$

$$PV = 3000 \left(\frac{0.32316}{0.05} \right)$$

$$PV = 3000(6.4632) = 19,389.60$$

$$\boxed{PV \approx \$19,389.60}$$

Net present value:

$$NPV = 19,389.60 - 12,000 = 7,389.60$$

$$\boxed{NPV \approx \$7,389.60}$$

Decision:

Because:

$$NPV > 0$$

Johny should invest.

$$\boxed{\text{Yes, Johny should invest}}$$